

Quantum Query Algorithms for the Constructive Diagonal Ramsey Theorem

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Abstract

The constructive diagonal Ramsey problem asks, given adjacency-oracle access to an N -vertex graph, for a clique or independent set of the order guaranteed by Ramsey's theorem. We give a bounded-error quantum algorithm that, for every $K \geq 2$ and $N \geq 4^{K-1}$, finds and verifies a homogeneous K -set using

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries with failure probability at most η . At the Ramsey scale $N = 2^n$, this yields a homogeneous set of order $\lfloor n/2 \rfloor + 1$ using $O(\sqrt{N} \log N \log(\log N/\eta))$ queries, improving on the $O(N)$ queries of the explicit Erdős–Szekeres recursion and giving, to our knowledge, the first sublinear worst-case algorithm for the Ramsey relation, whose query complexity was raised as a question by Jain, Li, Robere, and Xun (FOCS 2024). The bound also shows that the $N^{1-o(1)}$ quantum lower bound reported there cannot hold at its stated parameters; we show that the cited ingredients compose to an $\Omega(N^{1/12})$ lower bound instead.

The algorithm runs the constructive recursion over implicit candidate sets. Each set is represented by a short conjunction of adjacency constraints and sampled using capped unknown-solution quantum search, and a scale-aware concentration schedule balances estimation accuracy against the increasing cost of sampling deeper sets. We complement the upper bound with an $\Omega(N^{1-1/\sqrt{2}})$ randomized lower bound, transported from the random-Painter analysis of online Ramsey numbers, which holds on the uniform distribution $G(N, 1/2)$. On that distribution a greedy quantum search uses only $\tilde{O}(N^{1/4})$ queries, giving a provable polynomial quantum speedup for Ramsey search on random graphs. We also give an estimation-free size-biased recursion and extend it to every fixed number of edge colours.

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1 Introduction

Ramsey's theorem guarantees that every graph on $N \geq 4^{K-1}$ vertices contains a clique or an independent set of order K . Its classical constructive proof [8] repeatedly chooses a pivot and restricts the candidate set to one of the pivot's two colour neighbourhoods; truncating each retained neighbourhood to half its current size, it finds a homogeneous K -set with $O(N)$ adjacency queries. The corresponding total search problem, RAMSEY, has been studied in proof complexity and in TFNP [15, 16, 14, 20, 13]: the input graph on $N = 2^n$ vertices is given by a circuit or an oracle, and the target order is

model	input	lower bound	upper bound
randomized	worst case	$\Omega(N^{1-1/\sqrt{2}})$ [Proposition 4.1]	$O(N)$ [8]
randomized	$G(N, 1/2)$	$\Omega(N^{1-1/\sqrt{2}})$ [Proposition 4.1]	$\tilde{O}(N^{1/2})$ [Remark 4.4]
quantum	worst case	$\Omega(N^{1/12})$ [Proposition 4.5]	$\tilde{O}(N^{1/2})$ [Corollary 1.2]
quantum	$G(N, 1/2)$	open	$\tilde{O}(N^{1/4})$ [Corollary 4.3]

■ **Table 1** Bounded-error query complexity of finding a homogeneous set of order $\lfloor n/2 \rfloor + 1$ in a graph on $N = 2^n$ vertices, in the worst case and on the uniform distribution $G(N, 1/2)$; here $1 - 1/\sqrt{2} \approx 0.2929$. The lower bounds are the random-Painter estimate of [5] and the collision reduction of [13] at multiplicity two (Proposition 4.5 is stated for order $\lfloor n/2 \rfloor$ and hence valid here).

$K = n/2$. Jain, Li, Robere, and Xun [13, Section III] explicitly raise the question of the query complexity of this relation in the deterministic, randomized, and quantum models. This paper answers the quantum question up to a polynomial gap and clarifies the lower-bound record on both the quantum and the classical side.

The central observation is that the candidate sets of the constructive proof need not be materialized. After i rounds, membership in the current set is determined by the i adjacency constraints imposed by the previous pivots. These short predicates allow quantum search to sample from the same recursion at substantially smaller query cost. The threshold 4^{K-1} is the natural scale for this recursive construction.

1.1 Our results

We work in the promised valid-graph adjacency-oracle model. One coherent query returns the colour of a requested unordered vertex pair, and the output is a set of vertices inducing one colour. Our main result is the following.

► **Theorem 1.1** (Implicit-majority quantum Ramsey search). *Let $K \geq 2$, $N \geq 4^{K-1}$, and $0 < \eta < 1/2$. Given coherent edge-oracle access to a simple undirected graph on N vertices, there is a quantum algorithm which, with probability at least $1 - \eta$, outputs a K -clique or a K -vertex independent set using*

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries in the worst case. The algorithm verifies every returned witness and may output $\perp$ on the exceptional event.

At the succinct Ramsey scale, the theorem gives the following form.

► **Corollary 1.2.** *Let $n \geq 2$, $N = 2^n$, and $0 < \eta < 1/2$. Given coherent edge-oracle access to a simple graph on N vertices, one can find a clique or independent set of order $\lfloor n/2 \rfloor + 1$ using*

$$O\left(\sqrt{N} \log N \log \frac{\log N}{\eta}\right)$$

edge queries in the worst case.

67 **Proof.** Set $K = \lfloor n/2 \rfloor + 1$. Then $4^{K-1} = 2^{2\lfloor n/2 \rfloor} \leq 2^n = N$, so Theorem 1.1 applies,
 68 and $2^K \leq 2\sqrt{N}$, $K = O(\log N)$, and $\log(K/\eta) = O(\log(\log N/\eta))$. ◀

69 To our knowledge, this is the first sublinear worst-case query algorithm for the
 70 Ramsey relation in any model. Table 1 places it among the known bounds together
 71 with the further results of this paper.

72 An estimation-free recursion.

73 Following the colour of a uniformly sampled edge selects a colour class with probability
 74 proportional to its size, and a one-dimensional survival recurrence shows that this
 75 recursion completes with probability greater than $1/2$ from 4^{K-1} vertices. This
 76 gives an $O(2^K K^2 \log K \log(1/\eta))$ -query algorithm without any majority estimation
 77 (Proposition 3.2), which extends to every fixed number of edge colours (Corollary D.1).

78 A randomized lower bound.

79 Every randomized algorithm needs $\Omega(M^{1-1/\sqrt{2}})$ queries to find a homogeneous K -
 80 set in a graph on $M = 4^{K-1}$ vertices, even on the uniform distribution $G(M, 1/2)$
 81 (Proposition 4.1); this improves the $M^{1/4}$ black-box bound of [12, 14]. The proof
 82 transports the random-Painter weight estimate of [5] to the query model, with a
 83 self-contained version in Appendix C.

84 A quantum speedup on random graphs.

85 On $G(N, 1/2)$, a greedy quantum neighbourhood search finds a K -clique with $\tilde{O}(N^{1/4})$
 86 queries (Proposition 4.2). Together with the distributional form of the lower bound,
 87 this gives a polynomial separation between the quantum and randomized query
 88 complexities of the Ramsey relation on its natural hard distribution (Corollary 4.3).
 89 We are not aware of an earlier provable quantum speedup for this relation. In the
 90 worst case, no separation is claimed: the randomized complexity is only known to lie
 91 between $N^{1-1/\sqrt{2}}$ and N .

92 The quantum lower-bound record.

93 The paragraph of [13] that raises the query question also reports an $N^{1-o(1)}$ quantum
 94 lower bound at the default parameters, obtained by composing its collision reduction
 95 with the multicollision lower bound of [18]. Theorem 1.1 shows that no such bound can
 96 hold at these parameters, and Section 5 shows that the cited ingredients compose to
 97 an exponent of at most $1/12$ (at most $1/24$ under the parameter condition as printed),
 98 attained at multiplicity two. The resulting $\Omega(N^{1/12})$ bound (Proposition 4.5) is, to
 99 our knowledge, the best known quantum lower bound; the main theorems of [13] do
 100 not depend on the remark.

101 1.2 Techniques

102 Both algorithms represent the candidate set after round i by the conjunction of the
 103 adjacency constraints generated by the first i pivots, so a membership test costs

104 $O(i)$ edge queries, and sample it with capped unknown-solution search (Lemma 2.2);
 105 conditional on success, permutation symmetry makes each sample exactly uniform.
 106 The scale-aware recursion estimates a near-majority colour at every pivot. Sampling a
 107 deep set is exponentially more expensive than sampling an early one, so we distribute
 108 a fixed error budget across the levels, doubling the allowed error every six levels; a
 109 product invariant then keeps every candidate set within a constant factor of its ideal
 110 size, and the query sum is dominated by the last levels.

111 The lower bound and the random-graph speedup both rest on the predictable size
 112 of common neighbourhoods in $G(N, 1/2)$. A martingale weight argument in the style
 113 of [5] shows that an adaptive algorithm with T queries exposes a monochromatic K -set
 114 with probability at most $2 \cdot 2^{-\binom{K}{2} + c(c-1)} (2T)^{K-c}$ for every $c \leq K/2$, and optimizing
 115 c gives the exponent $2 - \sqrt{2}$; a union bound shows that every j -set has common
 116 neighbourhood of density about 2^{-j} , so greedy quantum search costs $O(2^{j/2})$ predicate
 117 evaluations at depth j .

118 Organization.

119 Section 2 fixes the query model and the sampler. Section 3 presents the two recursions,
 120 their analysis, and the proof sketch of Theorem 1.1; Section 4 states the lower bounds
 121 and the random-graph speedup. Section 5 discusses related work and the reported lower
 122 bound, and Section 6 lists open questions. All full proofs are in the appendices: the
 123 sampler and the encoding corollary in Appendix A, the upper bounds in Appendix B,
 124 the lower bounds and random-graph results in Appendix C, and the multicolour
 125 extension, numerical illustrations, and AI disclosure in Appendix D.

126 2 Oracle model and capped quantum sampling

127 Let $G: \binom{[N]}{2} \rightarrow \{0, 1\}$ be a promised simple undirected graph, extended symmetrically
 128 to ordered pairs with $G(u, u) = 0$. Its coherent edge oracle acts as $O_G|u, v, b\rangle =$
 129 $|u, v, b \oplus G(u, v)\rangle$. Query complexity counts calls to O_G ; reversible operations on
 130 vertex labels are free, as usual in the query model. The same asymptotic bound
 131 applies to the standard locally verifiable TFNP relation with arbitrary Boolean-circuit
 132 encodings.

133 ► **Corollary 2.1** (Locally verifiable arbitrary encoding). *The standard Ramsey TFNP*
 134 *relation that accepts either a local invalidity certificate or a locally verified homogeneous*
 135 *set has the same $O(2^K K \log(K/\eta))$ quantum-query upper bound as Theorem 1.1.*

136 The proof (Appendix A) canonicalizes the circuit into a simple graph, runs The-
 137 orem 1.1, and checks the $O(K^2)$ entries induced by the output for a self-loop or
 138 an asymmetric pair. Our algorithms use the following consequence of unknown-
 139 solution quantum search [2], proved in Appendix A; the analysis needs both bounded
 140 termination and exact conditional uniformity.

141 ► **Lemma 2.2** (Capped uniform marked-item search). *Let $T \subseteq [M]$ have density at least*
 142 *$\lambda > 0$, and suppose its membership predicate costs q edge queries. There is a quantum*
 143 *block which uses at most $O(q/\sqrt{\lambda})$ edge queries, succeeds with probability at least an*
 144 *absolute constant $p_0 > 0$, and, conditional on verified success, returns an exactly*

uniform element of T . The same predetermined cap makes the block terminate if the density promise is false or T is empty. Moreover, m independent uniform samples can be collected, except with probability δ , using a predetermined $O(m + \log(1/\delta))$ blocks whenever the density promise holds.

For comparison, the explicit classical recursion on $M = 4^{K-1} = 2^{r+1}$ vertices, $r = 2K - 3$, queries at level i all edges from a pivot to the other $s_i = M/2^i$ candidates and retains exactly $s_i/2$ vertices of a majority colour, for $\sum_{i < r} (s_i - 1) < 2M$ queries in total. The quantum algorithms retain the entire selected colour class but access it only implicitly.

3 Implicit-set recursions and the main theorem

3.1 An estimation-free size-biased recursion

Set $M = 4^{K-1}$ and $r = 2K - 3$, restrict the input to its first M vertices, let $S_0 = [M]$, and fix any $v_0 \in S_0$. At step $i = 0, \dots, r - 1$, define $T_i = S_i \setminus \{v_i\}$. Apply the repeated capped search from Lemma 2.2, with the universal density lower bound $1/M$, to sample an exactly uniform $x_i \in T_i$; abort if the fixed cap is exhausted. Query $c_i = G(v_i, x_i)$ and set

$$S_{i+1} = \{x \in T_i : G(v_i, x) = c_i\}, \quad v_{i+1} = x_i.$$

After the final step, put $w = v_r$. Among the r labels c_i , choose a colour that appears at least $K - 1$ times, and return the corresponding pivots together with w after directly verifying the induced edges. The algorithm never materializes S_i : its membership predicate is the conjunction of the edge constraints imposed by the earlier pivots, together with the exclusions of those pivots, and costs $O(i + 1)$ edge queries. Repeating each capped block $O(\log(2r))$ times reduces its conditional failure probability on a nonempty set to $O(1/r)$.

For integers $d, s \geq 0$, let $P_d(s)$ be the infimum, over all graphs, candidate sets of size s , and distinguished pivots in those sets, of the probability that the ideal size-biased recursion completes d additional steps; set $P_0(s) = 1$ for $s \geq 1$, $P_d(0) = 0$, and $P_d(1) = 0$ for $d \geq 1$.

► **Lemma 3.1** (Size-biased survival). *For every $d \geq 0$ and $s \geq 1$, $P_d(s) \geq (s - 2^d + 1)_+ / s$, where $(y)_+ = \max\{y, 0\}$. In particular, for $M = 4^{K-1}$ and $r = 2K - 3$, the ideal recursion completes with probability at least $1/2 + 1/M$.*

The proof is an induction on d : the sample selects the two colour classes of sizes $a + b = s - 1$ with probabilities $a/(s - 1)$ and $b/(s - 1)$, and $(x)_+ + (y)_+ \geq (x + y)_+$ collapses the resulting sum to one positive part. Exact conditional uniformity couples an implemented run to the ideal recursion, so charging each search failure probability $1/(100r)$ leaves completion probability at least 0.49; counting $O(\sqrt{M})$ tests per block, $O(\log(2r))$ blocks per level, and $O(i + 1)$ queries per test then gives the following bound (Appendix B).

► **Proposition 3.2** (Size-biased quantum recursion). *Under the hypotheses of Theorem 1.1, the estimation-free algorithm returns a verified homogeneous K -set with probability at least $1 - \eta$ using $O(K^2 2^K \log K \log(1/\eta))$ edge queries in the worst case.*

3.2 Scale-aware approximate majorities

The sharper recursion estimates a near-majority colour at each pivot. Fix $M = 4^{K-1}$ input vertices, $r = 2K - 3$, and $E = 1/16$. For $0 \leq i < r$, define

$$d_i = \left\lceil \frac{r-1-i}{6} \right\rceil, \quad z_i = 2^{-d_i}, \quad Z = \sum_{j=0}^{r-1} z_j,$$

and allocate

$$\varepsilon_i = E \frac{z_i}{Z}, \quad a_i = \frac{1}{2} - \varepsilon_i, \quad (1)$$

so that $\sum_i \varepsilon_i = E$. The allowable error doubles once every six levels, matching the increasing cost of sampling the deeper candidate sets. Before round i , the recorded pivots $v_0, \dots, v_{i-1}$ and colours $c_0, \dots, c_{i-1}$ define the exact implicit set

$$S_i = \{u \in [M] \setminus \{v_0, \dots, v_{i-1}\} : G(v_j, u) = c_j \text{ for every } j < i\}, \quad (2)$$

whose reversible membership predicate uses $O(i)$ edge queries.

At round i , use Lemma 2.2 to obtain a verified pivot $v_i \in S_i$. Then collect, with replacement,

$$m_i = \left\lceil \frac{1}{2\varepsilon_i^2} \log \frac{4r}{\eta} \right\rceil \quad (3)$$

independent uniform samples from $S_i \setminus \{v_i\}$, query their edges to v_i , and let $\hat{p}_i$ be the observed fraction of colour 1. Set $c_i = 1$ when $\hat{p}_i \geq 1/2$ and $c_i = 0$ otherwise; appending this edge constraint to Equation (2) defines S_{i+1} . After r rounds, one final capped search returns $w \in S_r$. Select a colour that occurs at least $K - 1$ times among $c_0, \dots, c_{r-1}$, take any $K - 1$ pivots carrying that colour, add w , query all $\binom{K}{2}$ induced edges, and return the set only if this check confirms homogeneity. Every search and batch has a predetermined cap, so every execution terminates; a failed search or batch returns $\perp$.

3.3 Analysis

► **Lemma 3.3** (Conditional near-majorities). *Except with probability at most $\eta/2$, every completed sampling batch reached after successful searches and an accurate prefix satisfies $|\hat{p}_i - p_i| \leq \varepsilon_i$, where p_i is the true colour-1 fraction from v_i to $S_i \setminus \{v_i\}$.*

Conditional on the classical history before level i and on batch success, the samples are independent and uniform on a fixed set, so Hoeffding's inequality and Equation (3) bound the inaccuracy probability by $\eta/(2r)$; a union bound over the first inaccurate level gives the lemma. Giving each of the $2r + 1$ searches and batches failure probability $\eta/(4r + 2)$ adds at most $\eta/2$, so all searches and estimates are good with probability at least $1 - \eta$. Let $s_i = |S_i|$, $A_0 = 1$, and $A_i = \prod_{j < i} a_j$.

► **Lemma 3.4** (Variable recurrence and density). *On every prefix satisfying Lemma 3.3, $s_i > MA_i - 1$ and $A_i \geq \frac{7}{8} 2^{-i}$. Consequently S_r is nonempty, $|S_i \setminus \{v_i\}|/M > \frac{3}{8} 2^{-i}$ for every sampling level $i < r$, and $|S_r|/M \geq 2^{-(r+1)} \geq \frac{3}{8} 2^{-r}$.*

221 An accurate majority retains at least a fraction a_i of the non-pivot vertices, so
 222 $s_{i+1} \geq a_i(s_i - 1)$; the bound on A_i follows from $\sum_i \varepsilon_i = 1/16$, the invariant propagates
 223 because $a_i < 1/2$, and $M2^{-r} = 2$ makes the integer s_r exceed $3/4$ (Appendix B).

224 ► **Lemma 3.5** (Exact homogeneous output). *Whenever the algorithm reaches its final*
 225 *verified output, that output is a K -clique or a K -vertex independent set.*

226 **Proof.** For $i < j$, the definition of S_{i+1} gives $G(v_i, v_j) = c_i$ and $G(v_i, w) = c_i$; a colour
 227 occurring $\lceil r/2 \rceil = K - 1$ times among the $r = 2K - 3$ colours therefore yields, with w ,
 228 a homogeneous K -set, and the final direct check is independent of earlier errors. ◀

229 **Proof sketch of Theorem 1.1.** With probability at least $1 - \eta$ all searches succeed
 230 and all estimates are accurate; then Lemma 3.4 guarantees every density promise used
 231 by the sampler and the nonemptiness of S_r , and Lemma 3.5 gives a homogeneous
 232 output. A capped block at level i uses $O(2^{i/2})$ membership tests of $O(i + 1)$ queries
 233 each, a batch uses $O(m_i)$ blocks, and $Z < 12$ gives $\varepsilon_i^{-2} = O(4^{d_i})$ in Equation (1).
 234 With $h = r - 1 - i$ and $4^{\lceil h/6 \rceil} \leq 4 \cdot 2^{h/3}$, the batches cost

$$235 \quad O\left(\log \frac{r}{\eta} \sum_{i < r} (i + 1) 2^{i/2} 4^{d_i}\right) = O\left(2^{r/2} \log \frac{r}{\eta} \sum_{h < r} (r - h) 2^{-h/6}\right) = O\left(r 2^{r/2} \log \frac{r}{\eta}\right),$$

236 the amplified searches cost the same order, and $r = 2K - 3$ gives $O(2^K K \log(K/\eta))$.
 237 The full computation is in Appendix B. ◀

238 4 Lower bounds and a quantum speedup on random graphs

239 4.1 A randomized classical lower bound

240 The following bound is the random-Painter estimate of Conlon, Fox, Grinshpun, and
 241 He [5, proof of Theorem 1] for online Ramsey numbers, transported to the query
 242 model. Because that estimate concerns uniformly random colourings, the bound holds
 243 on the uniform distribution $G(N, 1/2)$ and not only in the worst case.

244 ► **Proposition 4.1** (Randomized classical lower bound). *Let $K \geq 8$ and let $T =$*
 245 *$\lfloor 2^{(2-\sqrt{2})K-2} \rfloor$. For every $N \geq K$, every randomized algorithm that makes at most*
 246 *T edge queries outputs a homogeneous K -set of $G \sim G(N, 1/2)$ with probability at*
 247 *most $5/8$. Consequently, for $M = 4^{K-1}$, every randomized edge-query algorithm that*
 248 *finds a homogeneous K -set with worst-case probability at least $2/3$ uses $\Omega(2^{(2-\sqrt{2})K}) =$*
 249 *$\Omega(M^{1-1/\sqrt{2}})$ queries.*

250 The self-contained proof (Appendix C) tracks, through the matching number of the
 251 queried graph, the conditional probability that a vertex set becomes a monochromatic
 252 clique: a T -query algorithm fully exposes a monochromatic K -set with probability at
 253 most $2 \cdot 2^{-\binom{K}{2} + c(c-1)} (2T)^{K-c}$ for every integer $c \leq K/2$, which is at most $1/4$ at $c =$
 254 $\lfloor (1 - 1/\sqrt{2})K \rfloor$, and otherwise some pair of the output is an unexposed fair coin, so the
 255 success probability is at most $\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8}$. The exponent $1 - 1/\sqrt{2} \approx 0.2929$ improves
 256 the $M^{1/4}$ black-box bound of [12, 14]; no worst-case separation from Theorem 1.1 is
 257 claimed.

4.2 A quantum speedup on random graphs

► **Proposition 4.2** (Quantum greedy search on random graphs). *Let $K \geq 2$, $M = 4^{K-1}$, and $0 < \eta < 1/2$, and let $G \sim G(M, 1/2)$. There is a quantum algorithm which makes $O(K2^{K/2} \log(K/\eta))$ edge queries on every input, never outputs a set that is not a K -clique, and outputs a K -clique with probability at least $1 - \eta - \pi_K$ over G and its internal randomness, where $\pi_K = 2M^{K-1}e^{-2^{K-2}/12}$ satisfies $\pi_K < e^{-100}$ for all $K \geq 14$.*

With probability at least $1 - \pi_K$, every j -set with $j < K$ has at least $M2^{-j-1}$ common neighbours, and the algorithm then grows a clique one vertex at a time by the capped search of Lemma 2.2 with density parameter 2^{-j-1} at level j (Appendix C.3): no majority estimation is needed and the depth is K rather than $2K - 3$.

► **Corollary 4.3** (Separation on random graphs at the Ramsey scale). *Let $n \geq 26$, $N = 2^n$, $K = \lfloor n/2 \rfloor + 1$, $0 < \eta < 1/2$, and $G \sim G(N, 1/2)$.*

1. *A quantum algorithm making $O(N^{1/4} \log N \log(\log N/\eta))$ edge queries outputs a homogeneous K -set of G with probability at least $1 - \eta - \pi_K$.*
 2. *Every randomized algorithm making at most $\lfloor 2^{(2-\sqrt{2})K-2} \rfloor = \Omega(N^{1-1/\sqrt{2}})$ edge queries outputs a homogeneous K -set of G with probability at most $5/8$.*
- Since $1/4 < 1 - 1/\sqrt{2}$, the bounded-error quantum query complexity of the Ramsey relation on $G(N, 1/2)$ is polynomially smaller than its bounded-error randomized query complexity.*

The corollary applies Proposition 4.2 to the first $4^{K-1} \leq N$ vertices and Proposition 4.1 to $G(N, 1/2)$ (Appendix C.3).

► **Remark 4.4** (Classical greedy search). On the same expanding graphs, classical rejection sampling finds v_{j+1} with $\lceil 2^{j+1} \ln(K/\eta) \rceil$ trials of at most j queries each, except with probability η/K ; so for $G \sim G(M, 1/2)$ a classical greedy search outputs a verified K -clique with probability at least $1 - \eta - \pi_K$ using $O(K2^K \log(K/\eta)) = \tilde{O}(N^{1/2})$ queries, the classical entry of Table 1.

4.3 A quantum lower bound from collision finding

The reduction of Jain et al. [13, Lemma III.1], generalizing Komargodski, Naor, and Yaguev [14], maps a function $h: [N] \rightarrow [H]$ and a fixed graph A_0 on $[H]$ to the graph

$$A_h(u, v) = \mathbf{1}[h(u) = h(v)] \vee A_0(h(u), h(v)), \quad A_h(u, u) = 0, \quad (4)$$

on $[N]$. Composed with the quantum collision lower bound, it gives the best quantum lower bound we know; Section 5 compares it with the bound reported in [13]. A lower bound for target order $\lfloor n/2 \rfloor$ applies a fortiori to every larger target.

► **Proposition 4.5** (Quantum lower bound from collision finding). *Let $N = 2^n$ with $n \geq 8$. Every quantum algorithm that, given coherent edge-oracle access to a simple graph on N vertices, outputs a clique or independent set of order $\lfloor n/2 \rfloor$ with probability at least $2/3$ on every input makes $\Omega(N^{1/12})$ edge queries.*

In the proof (Appendix C.4), $H = 2^{\lfloor n/4 \rfloor}$ and A_0 is an Erdős graph on $[H]$ with no homogeneous set of order $2 \log_2 H$. Independent sets of A_h have distinct h -values

and map to independent sets of A_0 , cliques map to cliques, so every homogeneous set of order at least $2\log_2 H$ contains two vertices with the same h -value; a q -query Ramsey algorithm therefore finds a collision of a random h with $O(q + n)$ queries, which requires $\Omega(H^{1/3}) = \Omega(N^{1/12})$ [25].

5 Related work and the reported lower bound

Quantum algorithms for Ramsey problems.

Quantum approaches based on adiabatic computation, annealing, or quantum counting search for colourings that establish or test Ramsey-number bounds [9, 1, 24, 22, 23, 21]; our input is one fixed graph given by an oracle. Generic quantum algorithms for clique, independent set, and fixed-subgraph containment [6, 4, 17, 26] and quantum backtracking [19] operate in a closer oracle model; quantum walks also give sublinear-query algorithms for fixed sub-hypergraph containment [10]. We are not aware of an earlier implementation of the Erdős–Szekeres recursion over implicit sets in the adjacency-query model.

Constructive Ramsey search.

Ramsey search is connected to weak and structured pigeonhole principles [15, 16], its white-box hardness follows from collision-resistant hashing [14], and it belongs to the polynomial long-choice class PLC [20]. The closest matching query relation is studied in [13], which shows that it is not black-box reducible to PIGEON and raises the question of its query complexity in the deterministic, randomized, and quantum models. The online Ramsey game (see [5]), in which Builder queries edges and Painter answers adversarially, is the deterministic version of this question with an unbounded vertex set; its best known lower bound is the random-Painter bound used in Proposition 4.1.

Comparison with a reported quantum lower bound.

Model and parameter alignment. Definition II.6 of [13] uses $N = 2^n$ vertices, and the convention following Definition II.7 sets the default target to $K = n/2$. The unnumbered “Query complexity of RAMSEY” paragraph in Section III (p. 415) reports an $N^{1-o(1)}$ quantum lower bound, based on its Lemma III.1 and the multicollision lower bound of [18]. Lemma III.1 produces the product graph Equation (4), a simple graph whose coherent queries cost $O(1)$ queries to h , so the reduction lies within our promise; Corollary 2.1 covers arbitrary circuits, and Corollary 1.2 meets the default target after discarding one vertex. The two formulations therefore align.

Exponent obtained by direct substitution. Let H denote the multicollision range size and G the number of vertices in the resulting Ramsey instance. For fixed multiplicity t , the sufficient condition in Theorem I.3 of [13] is $G \geq H^{4t}/4^t$. At the boundary $G = \Theta_t(H^{4t})$, the exponent $\alpha_t = (2^{t-1} - 1)/(2^t - 1)$ of [18] transfers to G as $\alpha_t/(4t) = (2^{t-1} - 1)/(4t(2^t - 1)) \leq 1/24$ for $t \geq 2$, with equality at $t = 2$; for $t \geq 3$, $\alpha_t < 1/2$ gives $\alpha_t/(4t) < 1/(8t) \leq 1/24$. Increasing G relative to H only reduces the transferred exponent, and the multicollision theorem is stated for fixed t , so the cited

result does not support a growing-multiplicity limit either. The printed condition is not tight: the proof of Lemma III.1 only uses that a homogeneous set of the product graph maps to at most $K_0 - 1$ values of h , where $K_0 = 2 \log_2 H$ is the forbidden order in A_0 , so a homogeneous set of order $(t - 1)(K_0 - 1) + 1$ already forces a t -collision, the reduction only needs $G = \Theta_t(H^{4(t-1)})$, and the transferred exponent improves to $\alpha_t/(4(t - 1))$, which is $1/12$ at $t = 2$ and below $1/(8(t - 1)) \leq 1/16$ for $t \geq 3$. Direct composition of the cited ingredients therefore yields an exponent at most $1/12$, rather than $1 - o(1)$; Proposition 4.5 is exactly this $t = 2$ bound, and Theorem 1.1 rules out an $N^{1-o(1)}$ lower bound at these parameters. The remark is not used elsewhere in [13], and the black-box separation theorems of that paper are unaffected.

6 Discussion and open questions

The query improvement rests on storing each candidate set as the conjunction of the adjacency constraints created by earlier pivots, and on allocating estimation error according to the cost of each level. Table 1 leaves two central questions.

A worst-case quantum advantage.

Can the quantum and randomized query complexities be separated in the worst case? The randomized complexity on $N = 2^n$ vertices lies between $\Omega(N^{1-1/\sqrt{2}})$ and $O(N)$, while the quantum upper bound is $\tilde{O}(\sqrt{N})$. A randomized lower bound of $N^{1/2+\Omega(1)}$ would establish such a separation. The implicit-set recursion explains the difference between the available upper bounds: classical rejection sampling at depth i needs 2^i membership tests, whereas quantum search needs $2^{i/2}$.

The optimal quantum exponent.

Is $\sqrt{N}$ optimal in the worst case? The current quantum bounds are $\Omega(N^{1/12})$ and $O(\sqrt{N} \log N \log \log N)$. The final levels of the recursion search sets of constant size inside a universe of size $\Theta(N)$, but these sets are defined by $O(\log N)$ adjacency constraints, so the optimality of Grover search does not establish a lower bound for this relation.

Scope.

The distributional separation on $G(N, 1/2)$ in Corollary 4.3 and the multicolour bound in Corollary D.1 hold under their stated contracts. Appendix D.2 reports small sanity-check experiments; the query bounds rest on the proofs in the appendices. The paper claims query bounds only, not gate complexity or hardware performance.

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A Proofs of the sampler lemma and the encoding corollary

Proof of Lemma 2.2. Begin in the uniform superposition on $[M]$ and apply the randomized-iteration unknown-solution search of [2]. When the marked density is at least λ , truncate the geometric iteration schedule once its maximum iteration count reaches a sufficiently large constant multiple of $1/\sqrt{\lambda}$. The cited analysis gives a success probability $p_0 > 0$, independent of M and λ , using $O(1/\sqrt{\lambda})$ calls to the membership predicate. Because the cap is fixed before any oracle outcome is observed, the block terminates for every predicate.

For each fixed number of Grover iterations, all marked basis states have the same amplitude. Randomizing the iteration count preserves this permutation symmetry. After measuring a candidate, evaluate the exact membership predicate and accept only a marked outcome. Conditional on acceptance, the output is therefore uniform on the marked set T . Independent blocks reinitialize all registers and random choices, so their accepted outputs are independent conditional on the block-success indicators.

To collect a batch, run

$$B = C(m + \log(1/\delta))$$

independent capped blocks, where the absolute constant C is chosen from p_0 . The number of accepted blocks stochastically dominates $\text{Bin}(B, p_0)$. A Chernoff lower-tail bound [3] makes the probability of fewer than m acceptances at most δ . Conditional on any success pattern with at least m acceptances, the first m accepted outputs are independent and uniform on T . ◀

Proof of Corollary 2.1. Given a circuit $C(u, v)$, define the canonical simple graph

$$A(u, v) = \begin{cases} C(\min\{u, v\}, \max\{u, v\}), & u \neq v, \\ 0, & u = v. \end{cases}$$

One query to A uses $O(1)$ queries to C , including uncomputation. Run Theorem 1.1 on A and let W be the returned homogeneous set. Query the diagonal and both ordered entries induced by W , using $O(K^2)$ additional queries. A self-loop or an asymmetric pair is a local invalidity certificate. Otherwise C agrees with A on W , so W is a valid clique or independent-set witness. The verification cost is absorbed by the bound in Theorem 1.1. ◀

B Proofs for Section 3

B.1 The size-biased recursion

Proof of Lemma 3.1. The case $d = 0$ is immediate. Suppose $d \geq 1$ and $s \geq 2$, and let the two colour classes among the $s - 1$ non-pivot vertices have sizes a and b . A uniform sample selects these classes with probabilities $a/(s - 1)$ and $b/(s - 1)$. The induction hypothesis therefore gives

$$P_d(s) \geq \frac{aP_{d-1}(a) + bP_{d-1}(b)}{s - 1} \geq \frac{(a - 2^{d-1} + 1)_+ + (b - 2^{d-1} + 1)_+}{s - 1}.$$

Because $(x)_+ + (y)_+ \geq (x+y)_+$ and $a+b = s-1$, the numerator is at least $(s-2^d+1)_+$. Replacing the denominator $s-1$ by s weakens the bound and proves the claim; the definitions cover $s \leq 1$. Finally, $M = 2^{r+1}$, so the bound at $(d, s) = (r, M)$ is $(M - 2^r + 1)/M = 1/2 + 1/M$. ◀

Exact conditional uniformity of each successful capped search couples an implemented run to the ideal recursion. Assign each of the r searches failure probability at most $1/(100r)$. A union bound decreases the completion probability by at most $1/100$, so a single fixed-cap run succeeds with probability at least 0.49 . Independent repetition, followed by exact verification, amplifies this probability to $1 - \eta$ using $O(\log(1/\eta))$ runs. On every completed run, nesting gives $G(v_i, v_j) = G(v_i, w) = c_i$ for $j > i$; the pigeonhole argument in Lemma 3.5 then gives a homogeneous output.

Proof of Proposition 3.2. A capped sample from a nonempty subset of the M -element universe costs $O(\sqrt{M})$ membership tests. Repeating the block $O(\log(2r))$ times reduces its failure probability to $O(1/r)$. Since the membership predicate at level i costs $O(i+1)$ edge queries, one fixed-cap run uses

$$O\left(\sqrt{M} \log(2r) \sum_{i=0}^{r-1} (i+1)\right) = O(K^2 2^K \log K)$$

edge queries. The survival analysis above gives a constant success probability for each run, including capped-search failures and final verification. Repeating independent runs $O(\log(1/\eta))$ times proves the stated worst-case bound. ◀

B.2 The scale-aware recursion

Proof of Lemma 3.3. Condition on the complete classical history $\mathcal{H}_i$ before the samples at level i . The graph, S_i , and v_i are then fixed. Let $\mathcal{B}_i$ be the event that the verified batch succeeds. Conditional on $\mathcal{B}_i$, Lemma 2.2 gives independent uniform samples from the fixed set. Hoeffding's inequality [11] and Equation (3) imply

$$\Pr(|\hat{p}_i - p_i| > \varepsilon_i \mid \mathcal{H}_i, \mathcal{B}_i) \leq 2e^{-2m_i \varepsilon_i^2} \leq \frac{\eta}{2r}.$$

Multiplying by $\Pr(\mathcal{B}_i \mid \mathcal{H}_i) \leq 1$ gives the same upper bound for a completed but inaccurate batch conditional on $\mathcal{H}_i$. Apply this bound to the first inaccurate completed level after an accurate prefix. The tower property removes the conditioning, and a union bound over the r possible levels completes the proof. The argument does not require independence across levels. ◀

There are r pivot searches, r sampling batches, and one final search. Give each operation failure probability at most $\eta/(4r+2)$ using the capped amplification in Lemma 2.2. Applying these conditional bounds after good prefixes, as above, shows that their union has probability at most $\eta/2$. Together with Lemma 3.3, all searches and estimates used below are good with probability at least $1 - \eta$.

Proof of Lemma 3.4. An accurate empirical majority retains at least the fraction $a_i = 1/2 - \varepsilon_i$ of the non-pivot vertices. Hence

$$s_{i+1} \geq a_i(s_i - 1). \tag{5}$$

533 Since $\sum_i \varepsilon_i = 1/16$ and $\prod_j (1 - x_j) \geq 1 - \sum_j x_j$ for $x_j \in [0, 1]$,

$$534 \quad A_i = 2^{-i} \prod_{j < i} (1 - 2\varepsilon_j) \geq 2^{-i} \left(1 - 2 \sum_{j < i} \varepsilon_j \right) \geq \frac{7}{8} 2^{-i}.$$

535 We prove $s_i > MA_i - 1$ by induction. The claim holds at $i = 0$. If it holds at level
536 i , then Equation (5) and $a_i < 1/2$ give

$$537 \quad s_{i+1} > a_i(MA_i - 2) = MA_{i+1} - 2a_i > MA_{i+1} - 1.$$

538 Because $M2^{-r} = 2$, we obtain $s_r > (7/8)2 - 1 = 3/4$, and therefore the integer s_r is
539 positive. For $i < r$, we have $M2^{-i} \geq 4$ and

$$540 \quad s_i - 1 > MA_i - 2 \geq \frac{7}{8}M2^{-i} - 2 \geq \frac{3}{8}M2^{-i}.$$

541 Finally, $s_r \geq 1$ and $1/M = 2^{-(r+1)}$, which proves the last assertion. $\blacktriangleleft$

542 **Proof of Theorem 1.1.** Set $M = 4^{K-1}$ and run the algorithm of Section 3 on any
543 fixed set of M input vertices. The failure allocation above, together with Lemma 3.3,
544 implies that all required searches succeed and all completed estimates are accurate
545 with probability at least $1 - \eta$. On this event, Lemma 3.4 guarantees every density
546 promise used by the sampler and ensures that S_r is nonempty. The final output is
547 homogeneous by Lemma 3.5. Because the algorithm verifies the returned set directly,
548 every non- $\perp$ output is valid even outside the good event.

549 It remains to bound the number of queries. By Lemma 3.4, a capped search block
550 at level i uses $O(2^{i/2})$ membership tests. Computing and uncomputing membership
551 and querying the sampled edge costs $O(i + 1)$ edge queries. Define

$$552 \quad b_i = (i + 1)2^{i/2}.$$

553 The batch cap in Lemma 2.2 uses $O(m_i + \log(r/\eta)) = O(m_i)$ blocks. By Equation (3),
554 the total cost of the sampling batches is therefore

$$555 \quad O\left(\log \frac{r}{\eta} \sum_{i=0}^{r-1} b_i \varepsilon_i^{-2}\right). \tag{6}$$

556 At most six indices share any value of d_i , and hence

$$557 \quad Z = \sum_i 2^{-d_i} < 6 \sum_{d \geq 0} 2^{-d} = 12.$$

558 Using Equation (1),

$$559 \quad \sum_{i=0}^{r-1} b_i \varepsilon_i^{-2} = O\left(\sum_{i=0}^{r-1} (i + 1)2^{i/2} 4^{d_i}\right).$$

560 Write $h = r - 1 - i$. Since $4^{\lceil h/6 \rceil} \leq 4 \cdot 2^{h/3}$, the last display is at most

$$561 \quad O\left(2^{r/2} \sum_{h=0}^{r-1} (r - h)2^{-h/6}\right) = O(r2^{r/2}).$$

562 The amplified pivot searches and the final search cost

$$563 \quad O\left(\log \frac{r}{\eta} \sum_{i=0}^r b_i\right) = O\left(r 2^{r/2} \log \frac{r}{\eta}\right),$$

564 and the $O(K^2)$ output verification is smaller. Substituting $r = 2K - 3$ into Equation (6)
565 gives

$$566 \quad O\left(2^K K \log \frac{K}{\eta}\right),$$

567 as claimed. Each search begins in the uniform state on the fixed power-of-two universe
568 $[M]$, and the recorded pivot labels and colours define the reversible membership
569 predicate for S_i , so the algorithm requires no materialized representation of any
570 candidate set. ◀

571 **C Proofs for Section 4**

572 **C.1 A weight bound for adaptive queries to a random graph**

573 We record the random-Painter estimate of Conlon, Fox, Grinshpun, and He [5, proof
574 of Theorem 1] in the form used in this paper. We follow their weight argument, taking
575 the matching number of the queried graph as the potential; removing the critical edge
576 from a maximum matching then leaves a matching of the smaller set, which keeps the
577 recursion closed.

578 Throughout this subsection $V \geq 2$, and $G \sim G(V, 1/2)$ assigns independent uniform
579 colours in $\{0, 1\}$ to the pairs of $[V]$. A deterministic algorithm queries T distinct
580 pairs adaptively. Let $e_1, \dots, e_T$ be the queried pairs, let Q_t be the graph formed by
581 $e_1, \dots, e_t$, and let $\mathcal{F}_t$ be the σ -algebra generated by the first t answers. Because the
582 algorithm is deterministic, e_{t+1} is an $\mathcal{F}_t$ -measurable pair distinct from $e_1, \dots, e_t$, and
583 its colour is therefore a uniform bit independent of $\mathcal{F}_t$.

584 For $U \subseteq [V]$ and $0 \leq t \leq T$, let $e_t(U)$ be the number of pairs inside U among
585 $e_1, \dots, e_t$, and call U *consistent at time t* if all of these pairs have colour 1. Define

$$586 \quad w(U, t) = \begin{cases} 2^{-\binom{|U|}{2} + e_t(U)}, & U \text{ is consistent at time } t, \\ 0, & \text{otherwise,} \end{cases}$$

587 the conditional probability, given $\mathcal{F}_t$, that U becomes a colour-1 clique when its
588 remaining pairs are revealed. Let $\nu(U, t)$ be the matching number of $Q_t[U]$; it is
589 nondecreasing in t and increases by at most one when a pair is added. For integers
590 $k \geq 2c \geq 0$ set

$$591 \quad w_{k,c}(t) = \sum_{|U|=k, \nu(U,t) \geq c} w(U, t).$$

592 ▶ **Lemma C.1 (Weight bound).** *For all integers $k \geq 2c \geq 0$,*

$$593 \quad \mathbb{E} w_{k,c}(T) \leq 2^{-\binom{k}{2} + c(c-1)} V^{k-2c} T^c.$$

Proof. Martingale property. Fix U . If $e_{t+1} \not\subseteq U$, then $w(U, t+1) = w(U, t)$. If $e_{t+1} \subseteq U$, then with probability $1/2$ the new pair has colour 1 and $w(U, t+1) = 2w(U, t)$, and otherwise $w(U, t+1) = 0$. Hence $\mathbb{E}[w(U, t+1) \mid \mathcal{F}_t] = w(U, t)$, and in particular $\mathbb{E} w(U, T) = w(U, 0) = 2^{-\binom{|U|}{2}}$. Summing over all k -sets gives the case $c = 0$:

$$\mathbb{E} w_{k,0}(T) = \binom{V}{k} 2^{-\binom{k}{2}} \leq V^k 2^{-\binom{k}{2}}.$$

Monotonicity. Since $\nu(U, \cdot)$ is nondecreasing, $w_{k,c}(t+1) \geq \sum_{\nu(U,t) \geq c} w(U, t+1)$, and each event $\{\nu(U, t) \geq c\}$ is $\mathcal{F}_t$ -measurable. Taking conditional expectations gives $\mathbb{E} w_{k,c}(t) \leq \mathbb{E} w_{k,c}(t+1)$.

Critical times. Let $c \geq 1$. For each U let $\tau(U) = \min\{t : \nu(U, t) \geq c\}$, with $\tau(U) = \infty$ if $\nu(U, T) < c$; this is a stopping time with $\tau(U) \geq 1$. Since $\{\tau(U) = t\} \in \mathcal{F}_t$ and $w(U, \cdot)$ is a martingale,

$$\mathbb{E} w_{k,c}(T) = \sum_{|U|=k} \sum_{t=1}^T \mathbb{E}[w(U, T) \mathbf{1}[\tau(U) = t]] = \sum_{t=1}^T \mathbb{E} w_{k,c}^*(t),$$

where

$$w_{k,c}^*(t) = \sum_{|U|=k, \tau(U)=t} w(U, t).$$

Charging. Fix t and write $e_t = \{x, y\}$. Suppose $|U| = k$ and $\tau(U) = t$. Then $\nu(U, t-1) = c-1$, $\nu(U, t) = c$, and $e_t \subseteq U$. Moreover, every maximum matching of $Q_t[U]$ contains e_t , since a maximum matching avoiding e_t would be a matching of size c in $Q_{t-1}[U]$. Fix such a matching M , let $M' = M \setminus \{e_t\}$, let A be the set of $2(c-1)$ vertices covered by M' , and let $U' = U \setminus \{x, y\}$. Then M' is a matching of $Q_{t-1}[U']$, so $\nu(U', t-1) \geq c-1$. We claim that at most $2(c-1)$ of the pairs between $\{x, y\}$ and U' belong to Q_{t-1} . Indeed, if $xa \in Q_{t-1}$ with $a \in U' \setminus A$, then $M' \cup \{xa\}$ is a matching of size c in $Q_{t-1}[U]$, a contradiction; the same holds for y . For each $ab \in M'$, the pairs xa and yb cannot both lie in Q_{t-1} , since $(M' \setminus \{ab\}) \cup \{xa, yb\}$ would again be a matching of size c in $Q_{t-1}[U]$; likewise xb and ya cannot both lie in Q_{t-1} . Hence at most two of xa, xb, ya, yb lie in Q_{t-1} , and the claim follows by summing over the $c-1$ edges of M' .

Consequently $e_t(U) \leq e_{t-1}(U') + 1 + 2(c-1)$, and U is consistent at time t only if U' is consistent at time $t-1$ and e_t has colour 1. Using $\binom{k}{2} - \binom{k-2}{2} = 2k-3$, we obtain

$$w(U, t) \leq 2^{-(2k-2c-2)} w(U', t-1) \quad \text{if } e_t \text{ has colour 1,}$$

and $w(U, t) = 0$ otherwise. The map $U \mapsto U'$ is injective because $U = U' \cup e_t$. Therefore

$$w_{k,c}^*(t) \leq \mathbf{1}[e_t \text{ has colour 1}] 2^{-(2k-2c-2)} w_{k-2,c-1}(t-1).$$

The indicator is independent of $\mathcal{F}_{t-1}$ with mean $1/2$, while $w_{k-2,c-1}(t-1)$ is $\mathcal{F}_{t-1}$ -measurable. Taking expectations and using monotonicity,

$$\mathbb{E} w_{k,c}^*(t) \leq 2^{-(2k-2c-1)} \mathbb{E} w_{k-2,c-1}(T),$$

and summing over $t \leq T$ gives

$$\mathbb{E} w_{k,c}(T) \leq T 2^{-(2k-2c-1)} \mathbb{E} w_{k-2,c-1}(T).$$

Iteration. Applying the last inequality c times, the exponents $2(k-2i) - 2(c-i) - 1$ for $i = 0, \dots, c-1$ sum to $2kc - 3c^2$, and the case $c = 0$ gives

$$\mathbb{E} w_{k,c}(T) \leq T^c 2^{-(2kc-3c^2)} V^{k-2c} 2^{-\binom{k-2c}{2}} = 2^{-\binom{k}{2}+c(c-1)} V^{k-2c} T^c,$$

because $2kc - 3c^2 + \binom{k-2c}{2} = \binom{k}{2} - c(c-1)$. $\blacktriangleleft$

► **Corollary C.2** (Fully exposed monochromatic sets). *Let $K \geq 2$, let $V \geq K$, let $T \geq 1$, and let a deterministic algorithm make at most T adaptive edge queries to $G \sim G(V, 1/2)$. For every integer $0 \leq c \leq K/2$, the expected number of K -sets all of whose $\binom{K}{2}$ pairs have been queried and carry the same colour is at most*

$$2 \cdot 2^{-\binom{K}{2}+c(c-1)} (2T)^{K-c}.$$

Proof. Simulate the algorithm while renaming vertices in order of first appearance in a query, using a table from original to canonical labels. This does not change the joint distribution of the queried graph and its colours, because the colours of distinct pairs are independent and uniform; the renamed algorithm queries pairs inside $[V']$ with $V' = \min\{V, 2T\}$, and the number of fully exposed monochromatic K -sets is unchanged. After the renamed algorithm halts, append distinct unqueried pairs inside $[V']$ in a fixed order until exactly $T' = \min\{T, \binom{V'}{2}\}$ pairs have been queried; this can only increase that number. A fully exposed colour-1 K -set U satisfies $\nu(U, T') = \lfloor K/2 \rfloor \geq c$ and $w(U, T') = 1$, so the number of such sets is at most $w_{K,c}(T')$. Lemma C.1 with $V' \leq 2T$ and $T' \leq T$ bounds its expectation by $2^{-\binom{K}{2}+c(c-1)} (2T)^{K-2c} T^c \leq 2^{-\binom{K}{2}+c(c-1)} (2T)^{K-c}$. The same bound holds for colour 0 by symmetry. $\blacktriangleleft$

C.2 Proof of the randomized lower bound

Proof of Proposition 4.1. Let $\beta = 2 - \sqrt{2}$ and $\alpha = 1 - 1/\sqrt{2}$, so that $\beta = 2\alpha$ and $\alpha^2 - 2\alpha + 1/2 = 0$. Set

$$T = \lfloor 2^{\beta K-2} \rfloor, \quad c = \lfloor \alpha K \rfloor = \alpha K + \epsilon, \quad -1 < \epsilon \leq 0.$$

Since $\alpha < 1/2$, we have $0 \leq c \leq K/2$, and $K \geq 8$ gives $T \geq 1$.

Consider first a deterministic algorithm making at most T queries to $G \sim G(N, 1/2)$. By Corollary C.2 and Markov's inequality, the probability that the queried pairs contain a fully exposed monochromatic K -set is at most

$$2 \cdot 2^{-\binom{K}{2}+c(c-1)} (2T)^{K-c}. \tag{7}$$

Since $2T \leq 2^{\beta K-1}$, the base-2 logarithm of Equation (7) is at most

$$1 - \binom{K}{2} + c(c-1) + (K-c)(\beta K-1) = 1 - \frac{K}{2} + \epsilon^2,$$

where the identity follows by substituting $c = \alpha K + \epsilon$ and $\beta = 2\alpha$ and using $\alpha^2 - 2\alpha + 1/2 = 0$. For $K \geq 8$ this quantity is at most -2 , so the event has probability at most $1/4$.

Now condition on the query transcript and suppose that the algorithm outputs a K -set whose internal pairs have not all been exposed in one colour. If the exposed internal pairs contain both colours, the output is not homogeneous. If they contain exactly one colour, at least one unexposed pair must match that colour, which has conditional probability at most $1/2$, because unexposed pairs are independent fair coins given the transcript. If no internal pair has been exposed, the success probability is $2^{1-\binom{K}{2}} \leq 1/2$ for $K \geq 3$. Thus the conditional success probability is at most $1/2$ outside the fully exposed event, and the success probability under $G(N, 1/2)$ is at most

$$\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8}.$$

A randomized algorithm making at most T queries is a distribution over such deterministic algorithms, so its success probability on $G \sim G(N, 1/2)$ is also at most $5/8$. Finally, an algorithm whose worst-case success probability on graphs with M vertices is at least $2/3 > 5/8$ must make more than T queries, and $M = 4^{K-1}$ gives

$$2^{\beta K} = \Theta(M^{\beta/2}) = \Theta(M^{1-1/\sqrt{2}}),$$

which proves the proposition. $\blacktriangleleft$

C.3 Proof of the random-graph speedup

Proof of Proposition 4.2. Let $M = 4^{K-1}$. For $W \subseteq [M]$ let

$$C(W) = \{u \in [M] \setminus W : G(w, u) = 1 \text{ for every } w \in W\}$$

be the common neighbourhood of W . Call G *expanding* if $|C(W)| \geq M2^{-j-1}$ for every $j \in \{1, \dots, K-1\}$ and every j -set W ; for $W = \emptyset$ the bound $|C(\emptyset)| = M$ holds trivially.

Random graphs are expanding. Fix $1 \leq j \leq K-1$ and a j -set W . Then $|C(W)| \sim \text{Bin}(M-j, 2^{-j})$ with mean $\mu = (M-j)2^{-j}$. Since $M = 4^{K-1} \geq 4(K-1) \geq 4j$, we have $\mu \geq \frac{3}{4}M2^{-j}$, hence $M2^{-j-1} \leq \frac{2}{3}\mu$ and $\mu \geq \frac{3}{4}M2^{-(K-1)} = \frac{3}{2} \cdot 2^{K-2}$. The Chernoff bound $\Pr[X \leq (1-\gamma)\mu] \leq e^{-\gamma^2\mu/2}$ [3] with $\gamma = 1/3$ gives

$$\Pr[|C(W)| < M2^{-j-1}] \leq e^{-\mu/18} \leq e^{-2^{K-2}/12}.$$

A union bound over the at most $\sum_{j=1}^{K-1} M^j \leq 2M^{K-1}$ sets W shows that G fails to be expanding with probability at most $\pi_K = 2M^{K-1}e^{-2^{K-2}/12}$. For the numerical claim, $M = 4^{K-1}$ gives

$$\ln \pi_K = (1 + 2(K-1)^2) \ln 2 - \frac{2^{K-2}}{12},$$

which equals $339 \ln 2 - 4096/12 < -106$ at $K = 14$, and $\ln \pi_{K+1} - \ln \pi_K = 2(2K-1) \ln 2 - 2^{K-2}/12 < 0$ for all $K \geq 14$, since the subtracted term doubles at each step while the first term grows by $4 \ln 2$. Hence $\pi_K < e^{-100}$ for all $K \geq 14$.

701 *The algorithm.* Set $W_0 = \emptyset$. For $j = 0, \dots, K-1$, given the clique $W_j =$
 702 $\{v_1, \dots, v_j\}$, run the capped search of Lemma 2.2 on $[M]$ for the predicate “ $u \notin W_j$
 703 and $G(v_i, u) = 1$ for all $i \leq j$ ”, with density parameter $\lambda_j = 2^{-j-1}$, repeating the block
 704 $O(\log(K/\eta))$ times so that its failure probability is at most η/K whenever the density
 705 promise holds. The predicate costs $O(j+1)$ edge queries, including uncomputation.
 706 Let v_{j+1} be the verified output and $W_{j+1} = W_j \cup \{v_{j+1}\}$. If any search fails, output
 707 $\perp$; otherwise output W_K .

708 *Correctness and cost.* Every accepted vertex was verified to be adjacent to all earlier
 709 vertices, so a non- $\perp$ output is a K -clique for every input graph. If G is expanding, then
 710 the search at level j has marked density at least λ_j , and a union bound over the K
 711 searches shows that the algorithm outputs W_K with probability at least $1 - \eta$. Hence
 712 it succeeds with probability at least $1 - \eta - \pi_K$ over G and its internal randomness.
 713 Each capped block at level j uses $O(2^{j/2})$ predicate evaluations, so the total number
 714 of edge queries is

$$715 \quad O\left(\log \frac{K}{\eta} \sum_{j=0}^{K-1} (j+1)2^{j/2}\right) = O\left(K2^{K/2} \log \frac{K}{\eta}\right)$$

716 on every input, because every block has a predetermined cap. $\blacktriangleleft$

717 **Proof of Corollary 4.3.** The subgraph induced on the first $M = 4^{K-1} \leq N$ vertices
 718 is distributed as $G(M, 1/2)$, so the first part follows from Proposition 4.2, using
 719 $2^{K/2} \leq 2^{n/4+1/2}$ and $K = O(\log N)$. The second part is the distributional statement
 720 of Proposition 4.1, which applies because $K \geq 8$. For the final sentence, note that
 721 $n \geq 26$ gives $K \geq 14$, so $\pi_K < e^{-100}$; with $\eta = 1/4$ the quantum algorithm succeeds
 722 with probability greater than $2/3$, while $5/8 < 2/3$. $\blacktriangleleft$

723 C.4 Proof of the quantum lower bound

724 **Proof of Proposition 4.5.** Let $m = \lfloor n/4 \rfloor \geq 2$ and $H = 2^m$. By Erdős’s bound
 725 $R(k, k) > 2^{k/2}$ for $k \geq 3$ [7], there is a graph A_0 on $[H]$ with no clique or independent
 726 set of order $2m$; fix one and hardwire it, so that evaluating A_0 costs no queries. For
 727 $h: [N] \rightarrow [H]$ let A_h be the graph Equation (4). It is a simple graph, and one coherent
 728 query to A_h costs $O(1)$ queries to h .

729 Every homogeneous set S of A_h with $|S| \geq 2m$ contains two vertices with the same
 730 h -value. Indeed, if S is independent, then its h -values are distinct, because equal values
 731 force an edge, and $h(S)$ is an independent set of A_0 ; hence $|S| = |h(S)| \leq 2m - 1$,
 732 which is impossible. So S is a clique, $h(S)$ is a clique of A_0 , and $|h(S)| \leq 2m - 1 < |S|$.

733 Since $2m \leq \lfloor n/2 \rfloor$, a set of the order in the statement qualifies. Given a q -query
 734 algorithm $\mathcal{A}$ as in the statement, run it on A_h , evaluate h on the $\lfloor n/2 \rfloor$ returned
 735 vertices, and output two of them with equal h -value. This finds a collision of h with
 736 probability at least $2/3$ using $O(q+n)$ queries to h , for every h . Finding a collision in a
 737 uniformly random function $[N] \rightarrow [H]$ with $N \geq H$ requires $\Omega(H^{1/3})$ quantum queries
 738 [25, 18], so $q + O(n) = \Omega(2^{m/3})$. Finally $m \geq (n-3)/4$ gives $2^{m/3} \geq 2^{-1/4}N^{1/12}$, and
 739 $n = o(N^{1/12})$, so $q = \Omega(N^{1/12})$. $\blacktriangleleft$

D Multicolour extension, numerical illustrations, and AI disclosure

D.1 A multicolour extension of size-biased survival

Suppose every edge has one of $q \geq 2$ colours, and define

$$r = q(K - 2) + 1, \quad B_0 = 0, \quad B_d = 1 + q + \cdots + q^{d-1} = \frac{q^d - 1}{q - 1} \quad (d \geq 1).$$

Choose the power of two

$$L_q = 2^{\lceil \log_2(2B_r) \rceil},$$

so $2B_r \leq L_q < 4B_r$.

► **Corollary D.1** (Multicolour size-biased Ramsey search). *For $q, K \geq 2$, every q -edge-coloured complete graph on at least L_q vertices admits a bounded-error quantum algorithm that outputs a monochromatic K -clique using*

$$O\left(r^2 \sqrt{L_q} \log(2r) \log \frac{1}{\eta}\right)$$

colour-oracle queries in the worst case, where r , B_r , and L_q are as defined above.

Proof. Run the size-biased recursion of Section 3 for r rounds on the first L_q vertices, now following the colour of a uniformly sampled edge among q possibilities. Let $P_d^{(q)}(s)$ be the minimum probability that the ideal recursion completes d further rounds from a candidate set of size s and a fixed pivot. We claim that

$$P_d^{(q)}(s) \geq \frac{(s - B_d)_+}{s}$$

for all $d \geq 0$ and $s \geq 1$. The case $d = 0$ is immediate. For $d \geq 1$ and $s \geq 2$, let the q colour classes among the $s - 1$ non-pivot vertices have sizes $a_1, \dots, a_q$. The induction hypothesis gives

$$P_d^{(q)}(s) \geq \frac{\sum_{j=1}^q a_j P_{d-1}^{(q)}(a_j)}{s - 1} \geq \frac{\sum_{j=1}^q (a_j - B_{d-1})_+}{s - 1}.$$

Since $\sum_j a_j = s - 1$ and $B_d = 1 + qB_{d-1}$, the numerator is at least $(s - B_d)_+$. This proves the claimed bound after weakening the denominator from $s - 1$ to s . The preceding display also gives the stronger bound $(s - B_d)_+/(s - 1)$ whenever $d \geq 1$ and $s \geq 2$. Consequently, starting from $L_q \geq 2B_r$, the ideal recursion survives all r rounds with probability strictly greater than $1/2$.

A membership predicate at level i contains i colour constraints and costs $O(i + 1)$ oracle queries. Using the universal density bound $1/L_q$, one capped sample costs $O((i + 1)\sqrt{L_q})$ colour queries. Repeating the capped block $O(\log(2r))$ times makes each sampling failure probability $O(1/r)$, so one fixed-cap run costs

$$O\left(\sqrt{L_q} \log(2r) \sum_{i=0}^{r-1} (i + 1)\right) = O\left(r^2 \sqrt{L_q} \log(2r)\right).$$

The union bound preserves a positive constant success probability. Since $r = q(K - 2) + 1$, one of the q recorded colours occurs at least $\lceil r/q \rceil = K - 1$ times. The corresponding pivots and the final vertex form a monochromatic K -clique by the same nesting argument as in Lemma 3.5. Independent repetition and direct verification amplify the success probability to $1 - \eta$ at an additional $O(\log(1/\eta))$ factor. ◀

D.2 Small-instance numerical illustrations

Ideal-sampler recursion. We ran the scale-aware recursion at $K = 3$, $N = 16$, and $r = 3$ for 1,000 trials in each of five graph families. The error schedule was $(\varepsilon_0, \varepsilon_1, \varepsilon_2) = (1/64, 1/64, 1/32)$, with sample counts $m_i = \lceil \log(600)/(2\varepsilon_i^2) \rceil$, corresponding to a total estimation-failure budget of 0.01. Pivots were sampled uniformly, majority-estimation samples were independent with replacement, and ties selected colour 1. We replaced quantum search by ideal uniform sampling to isolate the combinatorial recursion.

The deterministic inputs were the complete graph, the empty graph, and the balanced complete bipartite graph $K_{8,8}$. For the random inputs, one $G(16, 1/2)$ graph was held fixed across all 1,000 trials, while the final family drew a fresh $G(16, 1/2)$ graph in each trial. Every returned triple was verified; exhaustion or a nonhomogeneous output counted as failure.

Graph family on 16 vertices	failed/trials	minimum final $ S_3 $
complete	0/1000	13
empty	0/1000	13
balanced complete bipartite $K_{8,8}$	0/1000	6
one fixed $G(16, 1/2)$	0/1000	2
fresh $G(16, 1/2)$ per trial	0/1000	2

All 5,000 trials returned homogeneous triples. The final column records the smallest observed value of $|S_3|$ in each family.

Isolated Grover sampler. We evolved a 16-dimensional state vector initialized in the uniform superposition, using marked-set sizes 1, 2, 3, 4, 8, 15. The measurement distribution was averaged over a uniformly chosen number of Grover iterations from zero through four. The probability of measuring a marked item ranged from 0.4026746749 to 0.5973253251. Conditional on a marked outcome, the largest deviation from the uniform distribution on marked items was zero to machine precision. This calculation isolates the permutation symmetry used in Lemma 2.2.

Finite split-tree survival. Minimizing the ideal size-biased survival probability over every admissible colour-class split gives exact finite-depth comparisons with the analytic bounds. For nine binary levels starting from 1,024 candidates, the minimum was approximately 0.5086135566, above the lower bound 513/1024. For four ternary levels starting from 128 candidates, the minimum was approximately 0.7155732457, above 88/127.

804 **D.3 AI Disclosure**

805 We used ChatGPT and Claude as auxiliary tools for literature search, brainstorming,
806 proof checking, drafting and editing, and building experiment scripts (Sections 1, 3–6,
807 and Appendices A–D). The authors determined the research direction and presentations,
808 evaluated and revised all AI-assisted material, and take full responsibility for the
809 correctness, originality, and final content of the paper, including every proof and
810 reference.